

Weekly Meeting 2

UFSD reading and first 4H-SiC TCAD steps

Prof. Prabhakar Palni group meeting

Aryan Bandyopadhyay
Integrated MSc Physics

Focus today: two theory questions and two first TCAD runs

Plan for today

1. What “pixel size” means in the Cartiglia 2014 paper.
2. How pixel size changes capacitance, noise, and timing.
3. Why the paper can prefer larger d while the book prefers smaller d .
4. The first two Victory Device runs I have made so far.
5. What I will do next.

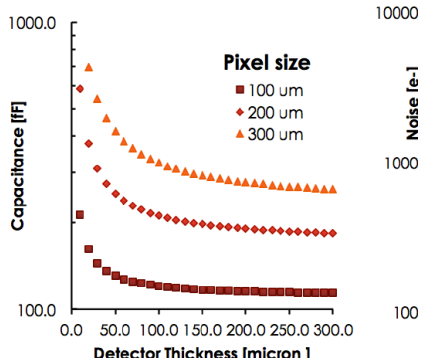
Scope This is still an early-stage update. The main progress is understanding the timing logic and getting the first 4H-SiC decks to run.

What pixel size means in this paper

- ▶ In Cartiglia 2014, l is the lateral pixel pitch, not the sensor thickness.
- ▶ The paper mainly compares $l = 100\ \mu\text{m}$ and $300\ \mu\text{m}$ pixels.
- ▶ For a square pixel, area scales as $A = l^2$.
- ▶ So when pixel pitch grows, the readout pad area also grows.

$$C_{\text{Det}} = \epsilon\epsilon_0 \frac{l^2}{d} + 0.2 \cdot 4l + 50\ \text{fF}$$

Back-plane term + neighbour term + stray term

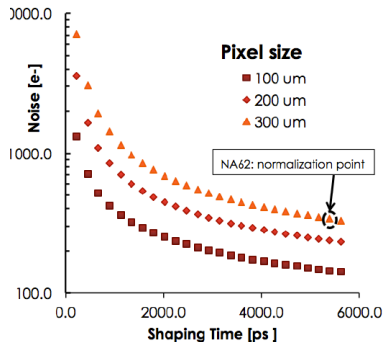


How capacitance enters noise and timing

$$N \propto \frac{C_{\text{Det}}}{\sqrt{t_r}}$$
$$\sigma_J = \frac{N}{dV/dt} = \frac{t_r}{S/N}$$

- ▶ Larger pixel $\Rightarrow$ larger C_{Det}
- ▶ Larger $C_{\text{Det}} \Rightarrow$ larger electronic noise
- ▶ Larger noise $\Rightarrow$ smaller S/N
- ▶ Smaller $S/N \Rightarrow$ worse jitter

Paper message Smaller pixels help timing mainly because they keep the detector capacitance and the front-end noise lower.



The paper also uses $V_{\text{th}} = 10N$, so larger noise also pushes the threshold upward.

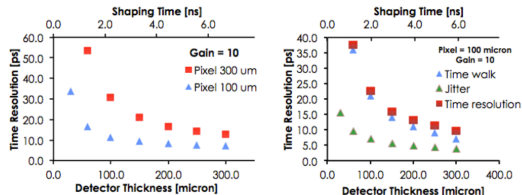
Why the paper can prefer larger d

$$\sigma_t^2 = \sigma_{TW}^2 + \sigma_J^2 + \sigma_{TDC}^2$$

$$\sigma_{TW} = \left[\frac{t_r V_{th}}{S} \right]_{RMS}, \quad \sigma_J = \frac{t_r}{S/N}$$

- ▶ Thicker detector $\Rightarrow$ more deposited charge $\Rightarrow$ larger signal S
- ▶ Larger S helps both time walk and jitter in this model
- ▶ Relative amplitude fluctuation also becomes smaller for larger d

So in this model If the main limits are S/N and fixed-threshold time walk, thicker can look better.



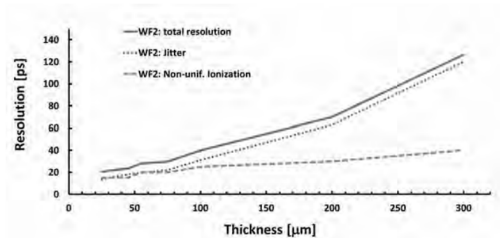
Cartiglia 2014: in this parametrization, the final timing trend improves as thickness increases.

Why the book can prefer smaller d

$$\frac{dI}{dt} \propto \frac{G}{d}$$

- ▶ The book gives more weight to intrinsic signal-shape variation, often called Landau noise.
- ▶ A thinner sensor gives a shorter collection path and a steeper rising edge.
- ▶ A thicker sensor gives more depth-dependent variation in the signal shape.

So the book is looking more at the intrinsic timing floor than at the older fixed-threshold S/N picture.



Book view: once gain and correction are good, thinner UFSDs can win because the intrinsic timing floor gets smaller.

Victory Device work so far

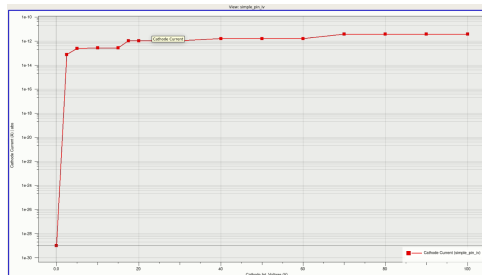
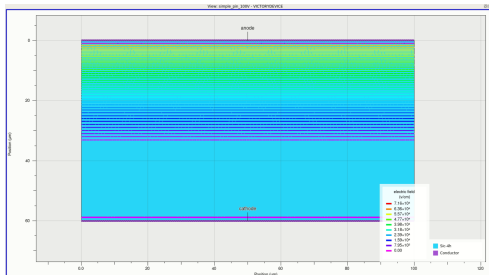
Deck	Purpose
PIN	baseline field and I-V check
LGAD-like	add a shallow high-field region

- ▶ Both runs are still first sanity checks.
- ▶ I am using them to check setup, bias sweep, and field localization.
- ▶ I am not treating them yet as calibrated detector results.

Current stage The main goal right now is to make the decks run in a controlled way and understand what each plot is actually telling me.

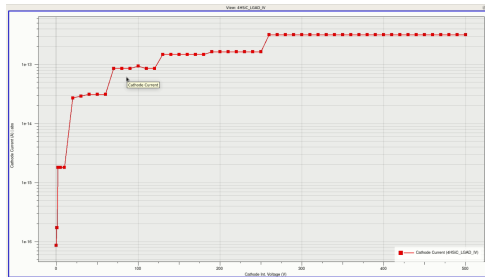
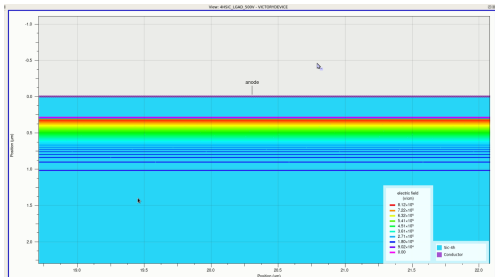
- ▶ PIN deck:
 - ▶ 60 μm vertical 4H-SiC PIN
 - ▶ SRH, BGN, fldmob, impact Nida
 - ▶ reverse sweep to 100 V
- ▶ LGAD-like deck:
 - ▶ 50 μm 4H-SiC structure
 - ▶ shallow top layer near $y = 0.3 \mu\text{m}$
 - ▶ SRH, Auger, fldmob, conmob, incomplete, impact Selb
 - ▶ reverse sweep to 500 V

First run: 4H-SiC PIN baseline check



- ▶ This was my first run, mainly to check geometry, contacts, and reverse bias.
- ▶ I only pushed the PIN case to 100 V in this first pass, so it is not yet a full high-bias study.
- ▶ The field pattern is mostly depth-dependent, which is expected for this simple vertical structure.

Second run: 4H-SiC LGAD-like deck



- ▶ This was my second run, where I moved from the PIN baseline to a first LGAD-like structure.
- ▶ The shallow top layer creates a localized high-field region near the anode, which is the main feature I wanted to see.
- ▶ The I-V curve is still a coarse first sweep, so I would not interpret the step-like jumps too strongly yet.
- ▶ At this stage, I would call this a first LGAD-like setup, not yet a calibrated gain result.

What these runs show, and what I still need to check

What is working

- ▶ I can build and run both a PIN deck and a first LGAD-like deck in Victory Device.
- ▶ I can extract I-V curves and visualize the electric field.
- ▶ The LGAD-like deck already shows field localization near the top junction.

What I still need to check

- ▶ Extend the PIN study to higher bias.
- ▶ Use a finer voltage sweep before reading too much into the I-V steps.
- ▶ Understand the contour banding carefully and separate plotting effects from physics.
- ▶ Tune the gain-layer profile and try to reproduce a known result.

Next: finish the remaining UFSD reading, learn the tool more carefully, and move toward one reproducible reference result.